

Applied Mathematics for Chemists

(화학수학)

Lecture 7

Multivariate Integral Calculus



Vector Derivatives

- Gradient (기울기) ∇f

scalar $\rightarrow$ vector

- Divergence (발산) $\nabla \cdot \vec{F}$

vector $\rightarrow$ scalar

- Curl (회전) $\nabla \times \vec{F}$

vector $\rightarrow$ vector

- Laplacian (라플라스) $\nabla^2 f$

scalar $\rightarrow$ scalar (2nd order)

Vector Derivatives

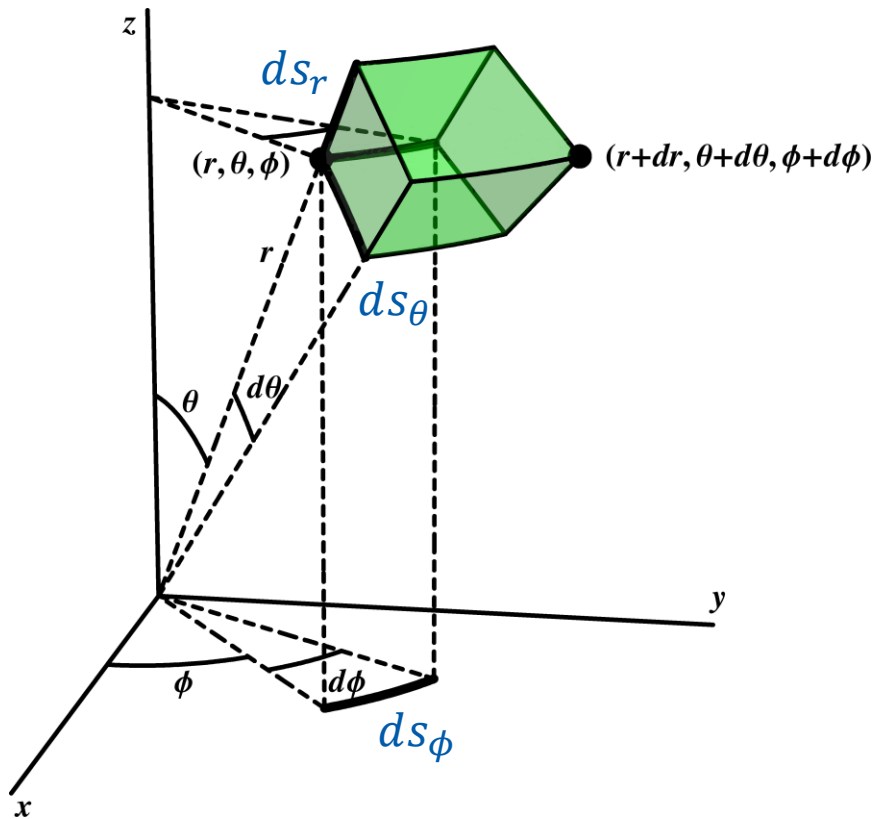
$$\nabla f = \hat{i} \left(\frac{\partial f}{\partial x} \right) + \hat{j} \left(\frac{\partial f}{\partial y} \right) + \hat{k} \left(\frac{\partial f}{\partial z} \right)$$

$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x} \right) + \left(\frac{\partial F_y}{\partial y} \right) + \left(\frac{\partial F_z}{\partial z} \right)$$

$$\nabla \times \vec{F} = \hat{i} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \hat{j} \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) + \hat{k} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

$$\nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2} \right) + \left(\frac{\partial^2 f}{\partial y^2} \right) + \left(\frac{\partial^2 f}{\partial z^2} \right)$$

Infinitesimal Displacements



$$ds_r = dr$$

$$ds_\theta = r d\theta$$

$$ds_\phi = r \sin \theta d\phi$$

Infinitesimal Displacements

In Cartesian coordinates,

$$d\vec{r} = \hat{i}dx + \hat{j}dy + \hat{k}dz$$

In spherical polar coordinates,

$$d\vec{r} = \hat{e}_r dr + \hat{e}_\theta r d\theta + \hat{e}_\phi r \sin \theta d\phi$$

In general,

$$d\vec{r} = \hat{e}_1 h_1 dq_1 + \hat{e}_2 h_2 dq_2 + \hat{e}_3 h_3 dq_3$$

Vector Derivatives in Other Coordinates

$$\nabla f = \hat{e}_1 \frac{1}{h_1} \frac{\partial f}{\partial q_1} + \hat{e}_2 \frac{1}{h_2} \frac{\partial f}{\partial q_2} + \hat{e}_3 \frac{1}{h_3} \frac{\partial f}{\partial q_3}$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(F_1 h_2 h_3)}{\partial q_1} + \frac{\partial(F_2 h_1 h_3)}{\partial q_2} + \frac{\partial(F_3 h_1 h_2)}{\partial q_3} \right]$$

$$\begin{aligned} \nabla \times \vec{F} = & \hat{e}_1 \frac{1}{h_2 h_3} \left[\frac{\partial(h_3 F_3)}{\partial q_2} - \frac{\partial(h_2 F_2)}{\partial q_3} \right] + \hat{e}_2 \frac{1}{h_1 h_3} \left[\frac{\partial(h_1 F_1)}{\partial q_3} - \frac{\partial(h_3 F_3)}{\partial q_1} \right] \\ & + \hat{e}_3 \frac{1}{h_1 h_2} \left[\frac{\partial(h_2 F_2)}{\partial q_1} - \frac{\partial(h_1 F_1)}{\partial q_2} \right] \end{aligned}$$

Vector Derivatives in Other Coordinates

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Line Integral (선적분)

Consider a differential with two independent variables:

$$dF = M(x, y)dx + N(x, y)dy$$

For the curve C joining (x_0, y_0) and (x_1, y_1) ,
the **line integral** (path integral) is defined as

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

Evaluation of Line Integral

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

The curve can be expressed as $x = x(y)$ or $y = y(x)$, so

$$\int_C dF = \int_C [M\{x, y(x)\}dx + N\{x(y), y\}dy]$$

$$\int_C dF = \int_{x_0}^{x_1} M\{x, y(x)\} dx + \int_{y_0}^{y_1} N\{x(y), y\} dy$$

Today's Class

- Line integrals
 - Exact and inexact differentials
 - Applications: thermodynamics
- Multiple integrals
 - Double and triple integrals
 - In different coordinate systems



Line Integrals of Exact Differentials

If dF is an exact differential, the line integral $\int_C dF$

- depends **only on the initial and final points**.
- equals to **$F(\text{final point}) - F(\text{initial point})$** .

For an exact differential dF of the function $F(x, y)$,
with the initial point (x_0, y_0) and the final point (x_1, y_1) ,

$$\int_C dF = \int_C \left[\left(\frac{\partial F}{\partial x} \right)_y dx + \left(\frac{\partial F}{\partial y} \right)_x dy \right] = F(x_1, y_1) - F(x_0, y_0)$$

Example 9.2

Calculate the line integral

$$\int_C dF = \int_C [(2x + 3y)dx + (3x + 4y)dy]$$

on the rectangular path from (0, 3) to (2, 3) and then to (2, 7)

Divide the path into C_1 from (0, 3) to (2, 3) and C_2 from (2, 3) to (2, 7)

$$\int_C dF = \int_{C_1} dF + \int_{C_2} dF$$

On C_1 , $y = 3$ (fixed) and $dy = 0$, so

$$\int_{C_1} [(2x + 3y)dx + (3x + 4y)dy] = \int_0^2 (2x + 9) dx = (x^2 + 9x) \Big|_0^2 = 22$$

Example 9.2

Calculate the line integral

$$\int_C dF = \int_C [(2x + 3y)dx + (3x + 4y)dy]$$

on the rectangular path from (0, 3) to (2, 3) and then to (2, 7)

Divide the path into C_1 from (0, 3) to (2, 3) and C_2 from (2, 3) to (2, 7)

$$\int_C dF = \int_{C_1} dF + \int_{C_2} dF = 22 + \int_{C_2} dF$$

On C_2 , $x = 2$ (fixed) and $dx = 0$, so

$$\int_{C_2} [(2x + 3y)dx + (3x + 4y)dy] = \int_3^7 (6 + 4y) dy = (6y + 2y^2) \Big|_3^7 = 104$$



The total value of the line integral is $22 + 104 = 126$.

Line Integrals of Inexact Differentials

If a differential is **not exact**, two line integrals on curves beginning and ending at the same points **will not necessarily yield the same result**.

Example 9.3

Show that the differential $du = dx + xdy$ is inexact, and carry out the line integral from $(0, 0)$ to $(2, 2)$ by two different paths:
path 1: the straight-line segment from $(0, 0)$ to $(2, 2)$; and
path 2: the rectangular path from $(0, 0)$ to $(2, 0)$ and then to $(2, 2)$.

If a differential $Mdx + Ndy$ is exact, $(\partial M/\partial y)_x = (\partial N/\partial x)_y$

$$(\partial M/\partial y)_x = (\partial(1)/\partial y)_x = 0$$

$$(\partial N/\partial x)_y = (\partial x/\partial x)_y = 1$$

Since $0 \neq 1$, the differential is **inexact**.

Example 9.3

Show that the differential $du = dx + xdy$ is inexact, and carry out the line integral from $(0, 0)$ to $(2, 2)$ by two different paths:
path 1: the straight-line segment from $(0, 0)$ to $(2, 2)$; and
path 2: the rectangular path from $(0, 0)$ to $(2, 0)$ and then to $(2, 2)$.

Consider the line integral on path 1. The formula for path 1 is $y = x$, so

$$\begin{aligned}\int_{C_1} du &= \int_{C_1} dx + \int_{C_1} x dy = \int_0^2 dx + \int_0^2 y dy \\ &= x \Big|_{x=0}^{x=2} + \frac{y^2}{2} \Big|_{y=0}^{y=2} = 2 + 2 = 4\end{aligned}$$

Example 9.3

Show that the differential $du = dx + xdy$ is inexact, and carry out the line integral from $(0, 0)$ to $(2, 2)$ by two different paths:
path 1: the straight-line segment from $(0, 0)$ to $(2, 2)$; and
path 2: the rectangular path from $(0, 0)$ to $(2, 0)$ and then to $(2, 2)$.

Now consider the line integral on path 2. The first segment is $y = 0$ and the second segment is $x = 2$.

$$\begin{aligned}\int_{C_2} du &= \int_{C_2} dx + \int_{C_2} x dy = \int_0^2 dx + \int_0^2 2 dy \\ &= x \Big|_{x=0}^{x=2} + 2y \Big|_{y=0}^{y=2} = 2 + 4 = 6\end{aligned}$$

Three Independent Variables

Consider

$$du = M(x, y, z)dx + N(x, y, z)dy + P(x, y, z)dz$$

The line integral on path C is

$$\int_C du = \int_C [M(x, y, z)dx + N(x, y, z)dy + P(x, y, z)dz]$$

Three Independent Variables

If C begins at (x_0, y_0, z_0) and ends at (x_1, y_1, z_1) ,

$$\begin{aligned}\int_C du &= \int_C [M(x, y, z)dx + N(x, y, z)dy + P(x, y, z)dz] \\ &= \int_{x_0}^{x_1} M[x, y(x), z(x)] dx + \int_{y_0}^{y_1} N[x(y), y, z(y)] dy \\ &\quad + \int_{z_0}^{z_1} P[x(z), y(z), z] dz\end{aligned}$$

Three Independent Variables

$$du = M(x, y, z)dx + N(x, y, z)dy + P(x, y, z)dz$$

If du is an exact differential,

$$\int_C du = \Delta u = u(x_1, y_1, z_1) - u(x_0, y_0, z_0)$$

Example 9.4

Consider the differential $du = yzdx + xzdy + xydz$.

Determine whether the differential is exact, and evaluate the line integral

$$\int_C du = \int_C (yzdx + xzdy + xydz)$$

on the path from $(0, 0, 0)$ to $(3, 3, 3)$ corresponding to $x = y = z$.

For $du = Mdx + Ndy + Pdz$, the exactness test is

$$\left(\frac{\partial M}{\partial y}\right)_{x,z} = \left(\frac{\partial N}{\partial x}\right)_{y,z}, \quad \left(\frac{\partial N}{\partial z}\right)_{x,y} = \left(\frac{\partial P}{\partial y}\right)_{x,z}, \quad \left(\frac{\partial M}{\partial z}\right)_{x,y} = \left(\frac{\partial P}{\partial x}\right)_{y,z}$$

and for the given differential,

$$z = z, \quad x = x, \quad y = y$$



so the differential is **exact**.

Example 9.4

Consider the differential $du = yzdx + xzdy + xydz$.

Determine whether the differential is exact, and evaluate the line integral

$$\int_C du = \int_C (yzdx + xzdy + xydz)$$

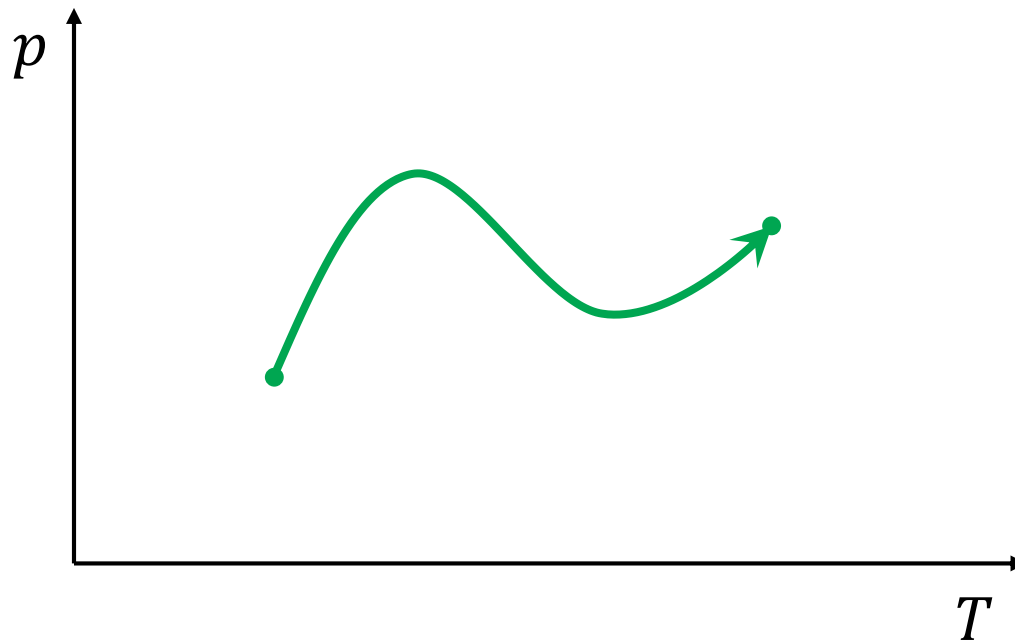
on the path from $(0, 0, 0)$ to $(3, 3, 3)$ corresponding to $x = y = z$.

Since the differential is exact and $u = xyz$,

$$\int_C du = u(3,3,3) - u(0,0,0) = 3 \times 3 \times 3 - 0 = 27$$

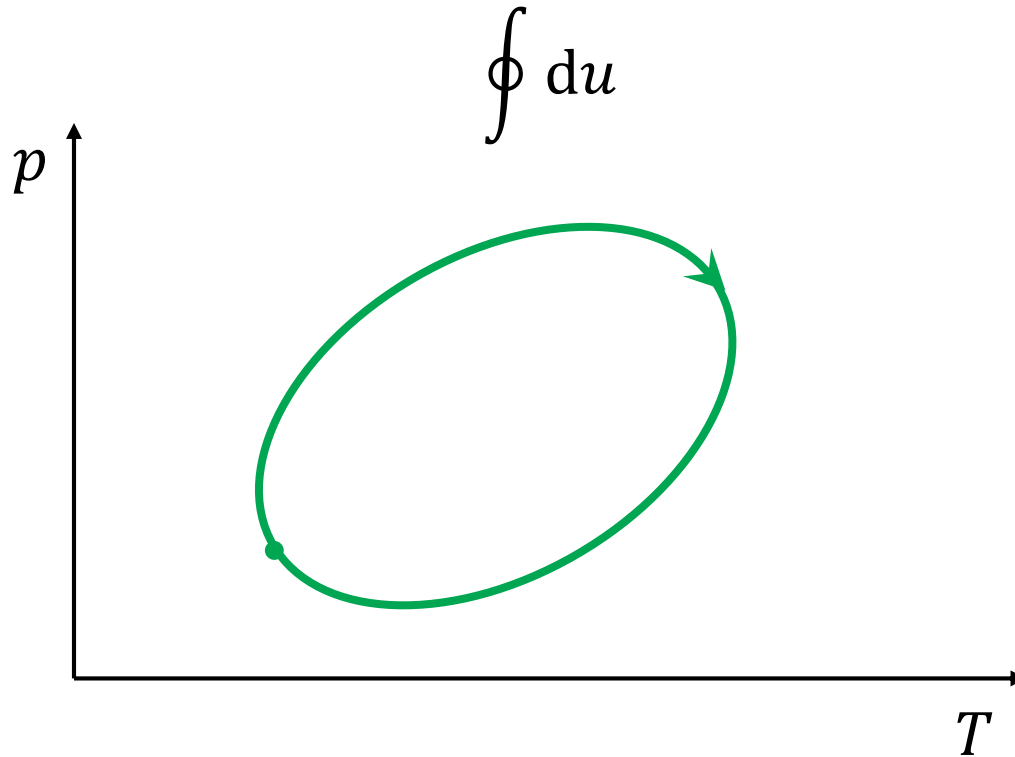
Line Integrals in Thermodynamics

- Equilibrium state: a point in a space
- Reversible process: a line integral



Cyclic Process (순환과정)

Process that begins and ends at the same state



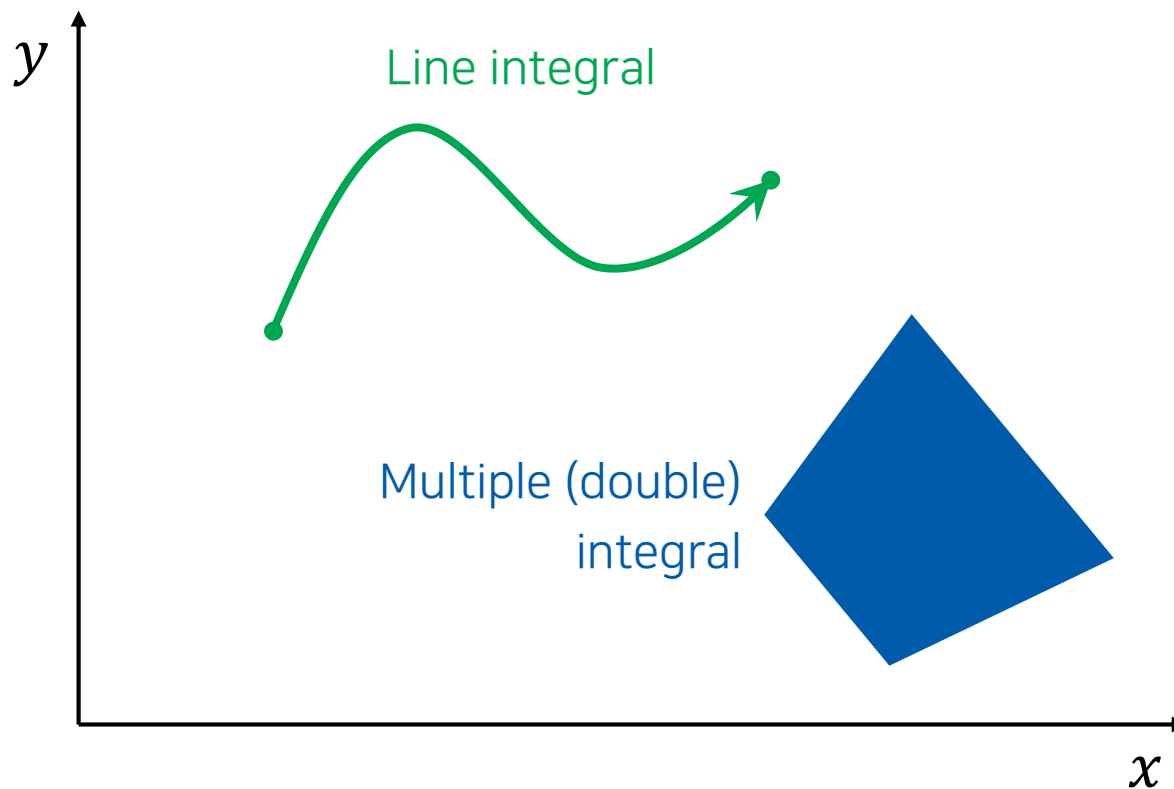
Cyclic Process (순환과정)

- If du is exact,

$$\oint du = 0$$

- *e.g.* $\oint dU = 0$ always
- If du is inexact, it is not necessary
 - *e.g.* $\oint \delta w \neq 0$ for some processes (see Example 9.5)

Line Integral vs. Multiple Integral



Double Integral (이중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} f(x, y) \, dy \, dx$$


- Carry out **the inner integration** first
 - Treat the other independent variable (x) as a constant
- Then carry out **the outer integration**

Example 9.6

Evaluate the double integral

$$I = \int_0^a \int_0^b (x^2 + 4xy) \, dy \, dx.$$

Carry out the inner integral first ($x = \text{constant}$):

$$\int_0^b (x^2 + 4xy) \, dy = [x^2y + 2xy^2]_0^b = x^2b + 2xb^2$$

Now carry out the outer integral:

$$I = \int_0^a (x^2b + 2xb^2) \, dx = \left[\frac{b}{3}x^3 + b^2x^2 \right]_0^a = \frac{a^3b}{3} + a^2b^2$$

Example 9.7

Evaluate the double integral

$$\int_0^a \int_0^{3x} (x^2 + 2xy + y^2) \, dy \, dx.$$

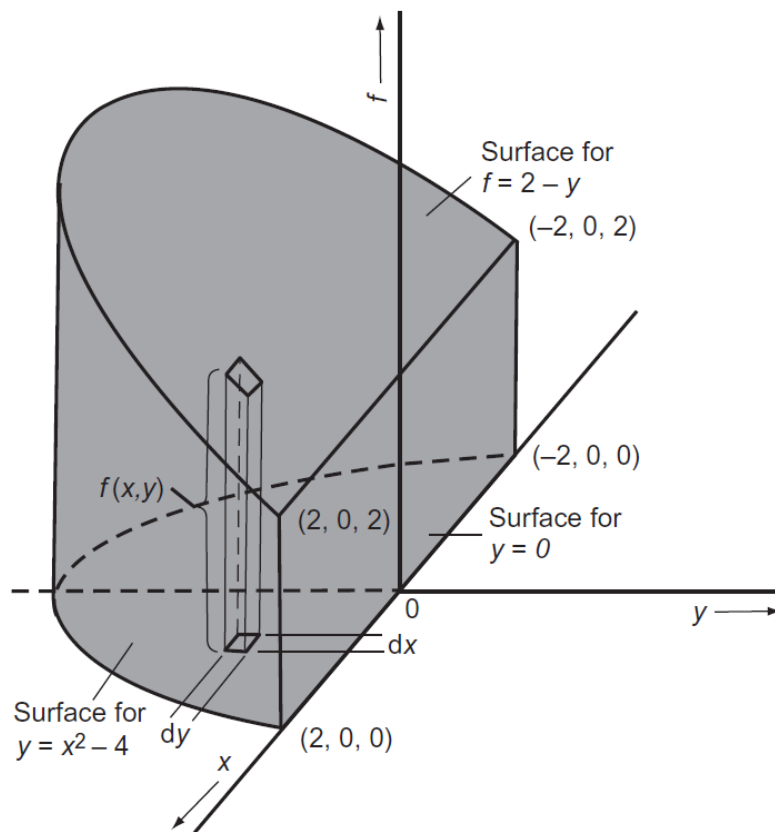
Carry out the inner integral first ($x = \text{constant}$):

$$\int_0^{3x} (x^2 + 2xy + y^2) \, dy = \left[x^2 y + xy^2 + \frac{y^3}{3} \right]_0^{3x} = 3x^3 + 9x^3 + 9x^3 = 21x^3$$

Now carry out the outer integral:

$$\int_0^a 21x^3 \, dx = \left[\frac{21}{4} x^4 \right]_0^a = \frac{21}{4} a^4$$

Application: Volume Calculation



$$V = \int_{-2}^2 \int_{x^2-4}^0 (2 - y) dy dx$$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.3)

Example 9.8

Evaluate the volume of the solid shown in the previous slide.

$$V = \int_{-2}^2 \int_{x^2-4}^0 (2-y) \, dy \, dx$$

Carry out the inner integral first ($x = \text{constant}$):

$$\int_{x^2-4}^0 (2-y) \, dy = \left[2y - \frac{y^2}{2} \right]_{x^2-4}^0 = -2(x^2-4) + \frac{(x^2-4)^2}{2} = \frac{x^4}{2} - 6x^2 + 16$$

Now carry out the outer integral:

$$\int_{-2}^2 \left(\frac{x^4}{2} - 6x^2 + 16 \right) dx = \left[\frac{x^5}{10} - 2x^3 + 16x \right]_{-2}^2 = \frac{192}{5}$$

Triple Integral (삼중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z) \, dz \, dy \, dx$$

- First, integrate z from c_1 to c_2 ($x, y = \text{constant}$)
- Then integrate y from b_1 to b_2 ($x = \text{constant}$)
- Lastly, integrate x from a_1 to a_2

Example 9.9

For integers n , m , and k , find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) dz dy dx.$$

Carry out the innermost integral first ($x, y = \text{constant}$):

$$\int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) dz = \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \int_0^c \sin^2\left(\frac{k\pi z}{c}\right) dz$$

From the integral table,

$$\int_0^c \sin^2\left(\frac{k\pi z}{c}\right) dz = \frac{c}{2}$$

Example 9.9

For integers n , m , and k , find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) dz dy dx.$$

Next, treating x as a constant,

$$\int_0^b \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \frac{c}{2} dy = \sin^2\left(\frac{n\pi x}{a}\right) \frac{c}{2} \int_0^b \sin^2\left(\frac{m\pi y}{b}\right) dy$$

Similarly, from the integral table,

$$\int_0^b \sin^2\left(\frac{m\pi y}{b}\right) dy = \frac{b}{2}$$

Example 9.9

For integers n , m , and k , find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) dz dy dx.$$

Lastly,

$$I = A^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) \frac{bc}{4} dx = \frac{A^2 bc}{4} \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx$$

Similarly, from the integral table,

$$\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{a}{2}$$

and we obtain $I = A^2 abc/8$.

Separation of Variables

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z) \, dz \, dy \, dx$$

If $f(x, y, z) = f_x(x)f_y(y)f_z(z)$,

$$I = \left[\int_{a_1}^{a_2} f_x(x) \, dx \right] \times \left[\int_{b_1}^{b_2} f_y(y) \, dy \right] \times \left[\int_{c_1}^{c_2} f_z(z) \, dz \right]$$

Example 9.9 Revisited

For integers n , m , and k , find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) dz dy dx.$$

The integral can be decomposed as

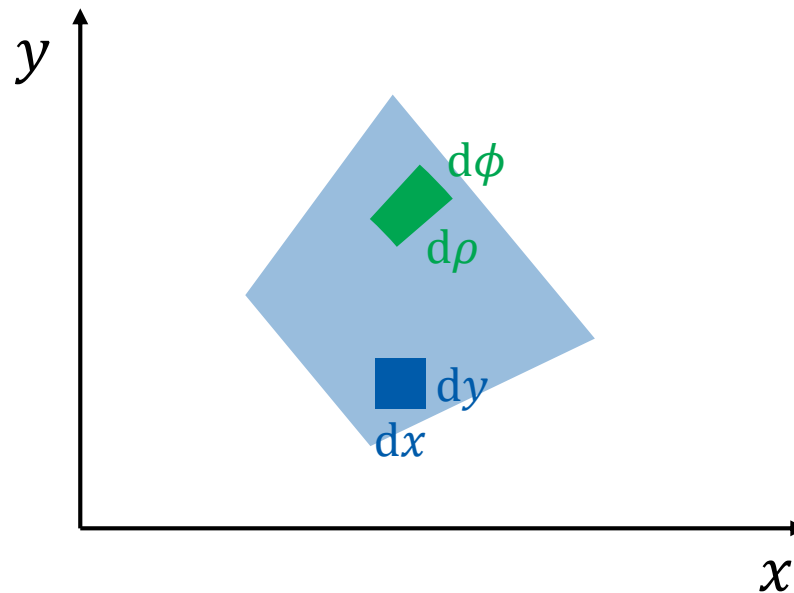
$$I = A^2 \left[\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx \right] \left[\int_0^b \sin^2\left(\frac{m\pi y}{b}\right) dy \right] \left[\int_0^c \sin^2\left(\frac{k\pi z}{c}\right) dz \right]$$

Hence, using the integral table,

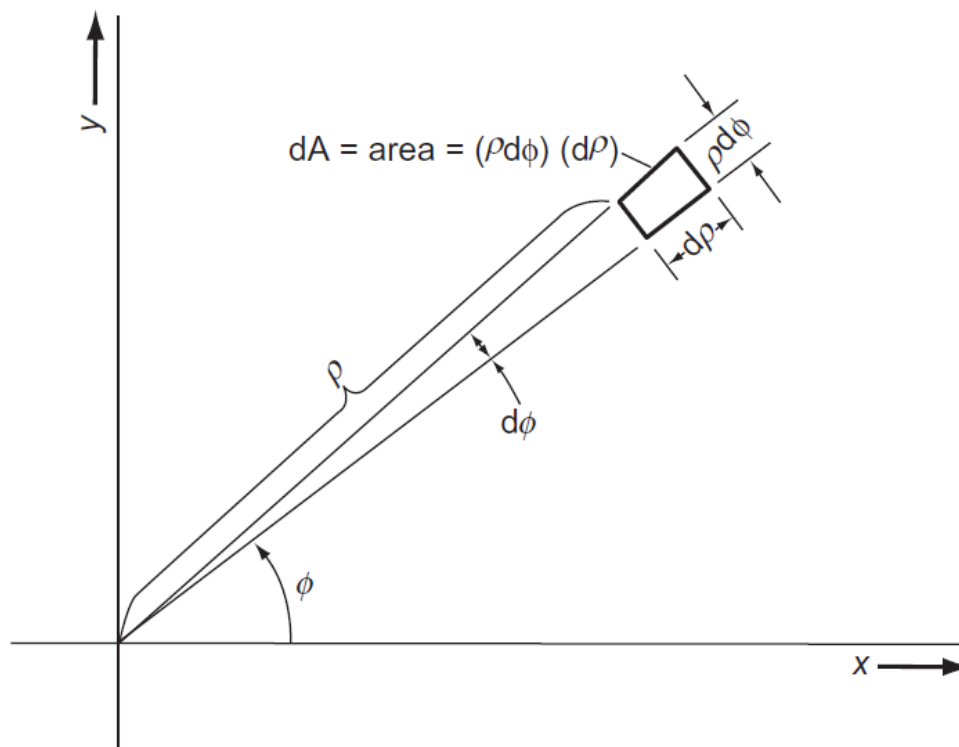
$$I = A^2 \times \frac{a}{2} \times \frac{b}{2} \times \frac{c}{2} = \frac{A^2 abc}{8}$$

Changing Variables in Double Integrals

$$\int \int f(x, y) \, dx \, dy \xrightarrow{?} \int \int f(\rho, \phi) \, d\rho \, d\phi$$



Method 1: Geometric Method



$$dA = dx dy = \rho d\rho d\phi$$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.4)

Conversion Formula

$$\iint f \, dA = \iint f(x, y) \, dx \, dy = \iint f(\rho, \phi) \rho \, d\rho \, d\phi$$

Example 9.10

$$\psi = B \exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x - y plane is equal to unity.

The transformation, $x = \rho \cos \phi$ and $y = \rho \sin \phi$, leads to

$$\begin{aligned}\psi &= B \exp[-a\{(\rho \cos \phi)^2 + (\rho \sin \phi)^2\}] \\ &= B \exp[-a\rho^2(\cos^2 \phi + \sin^2 \phi)] = Be^{-a\rho^2}\end{aligned}$$

Its square is

$$\psi^2 = B^2 e^{-2a\rho^2}$$

Example 9.10

$$\psi = B \exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x - y plane is equal to unity.

$$\psi^2 = B^2 e^{-2a\rho^2}$$

We require that

$$I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi^2 \, dx \, dy = 1$$

Substitution leads to

$$I = B^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-2a\rho^2} \, dx \, dy$$

Example 9.10

$$\psi = B \exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x - y plane is equal to unity.

$$\begin{aligned} I &= B^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-2a\rho^2} dx dy \\ &= B^2 \int_0^{2\pi} \int_0^{\infty} e^{-2a\rho^2} \rho d\rho d\phi \\ &= B^2 \int_0^{\infty} e^{-2a\rho^2} \rho d\rho \int_0^{2\pi} d\phi \\ I &= 2\pi B^2 \int_0^{\infty} e^{-2a\rho^2} \rho d\rho \end{aligned}$$

Example 9.10

$$\psi = B \exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x - y plane is equal to unity.

$$I = 2\pi B^2 \int_0^\infty e^{-2a\rho^2} \rho \, d\rho$$

Use the method of substitution, letting $u = 2a\rho^2$ and $du = 4a\rho d\rho$;

$$I = 2\pi B^2 \times \frac{1}{4a} \int_0^\infty e^{-u} \, du = \frac{\pi B^2}{2a}$$

Since $I = 1$, $B^2 = 2a/\pi$ and

$$B = \sqrt{2a/\pi}$$

Method 2: Jacobian

$$\int \int f(x, y) \, dx \, dy = \int \int f(\rho, \phi) \frac{\partial(x, y)}{\partial(\rho, \phi)} \, d\rho \, d\phi$$

where

$$\frac{\partial(x, y)}{\partial(\rho, \phi)} = \begin{vmatrix} \partial x / \partial \rho & \partial x / \partial \phi \\ \partial y / \partial \rho & \partial y / \partial \phi \end{vmatrix} = \left(\frac{\partial x}{\partial \rho} \right) \left(\frac{\partial y}{\partial \phi} \right) - \left(\frac{\partial x}{\partial \phi} \right) \left(\frac{\partial y}{\partial \rho} \right)$$

Jacobian Calculation

$$\frac{\partial(x, y)}{\partial(\rho, \phi)} = \left(\frac{\partial x}{\partial \rho} \right) \left(\frac{\partial y}{\partial \phi} \right) - \left(\frac{\partial x}{\partial \phi} \right) \left(\frac{\partial y}{\partial \rho} \right)$$

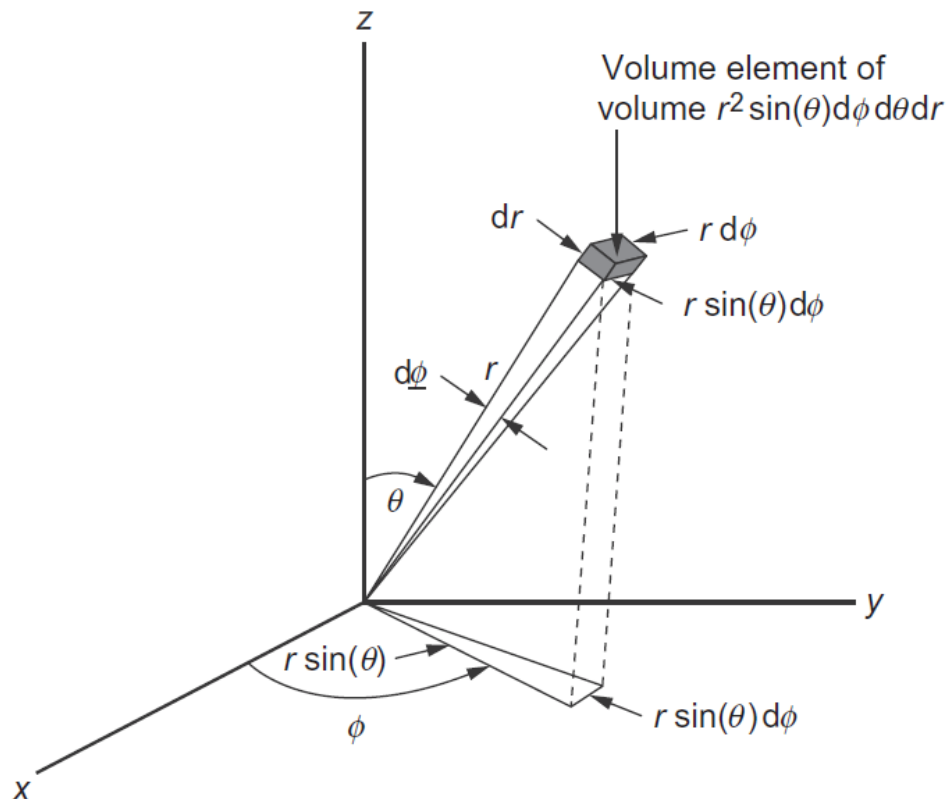
For $x = \rho \cos \phi$ and $y = \rho \sin \phi$,

$$\partial x / \partial \rho = \cos \phi, \quad \partial x / \partial \phi = -\rho \sin \phi$$

$$\partial y / \partial \rho = \sin \phi, \quad \partial y / \partial \phi = \rho \cos \phi$$

$$\text{Jacobian} = (\cos \phi)(\rho \cos \phi) - (-\rho \sin \phi)(\sin \phi) = \rho$$

Triple Integral in Spherical Coordinates



Jacobian: $r^2 \sin \theta$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.5)

Example 9.12

Evaluate the integral

$$I = \int_0^3 \int_0^\pi \int_0^{2\pi} r \cos^2 \phi \, r^2 \sin \theta \, d\phi \, d\theta \, dr.$$

The integral can be decomposed:

$$I = \int_0^3 r^3 \, dr \int_0^\pi \sin \theta \, d\theta \int_0^{2\pi} \cos^2 \phi \, d\phi$$

$$\int_0^3 r^3 \, dr = \left[\frac{1}{4} r^4 \right]_0^3 = \frac{81}{4}, \quad \int_0^\pi \sin \theta \, d\theta = [-\cos \theta]_0^\pi = 2$$

$$\int_0^{2\pi} \cos^2 \phi \, d\phi = \int_0^{2\pi} \frac{\cos(2\phi) + 1}{2} \, d\phi = \left[\frac{1}{4} \sin(2\phi) + \frac{1}{2} \phi \right]_0^{2\pi} = \pi$$



Hence, $I = 81\pi/2$.